

TEMPORAL DESCRIPTORS ASSOCIATED WITH PREDICTION ERRORS IN SUBCOOLED FLOW BOILING: EXPERIMENT-AWARE VALIDATION OF RANDOM FOREST MODELS

Magdalena PIASECKA*

*Department of Mechanics and Thermal Processes, Faculty of Mechatronics and Mechanical Engineering, Kielce University of Technology, al. Tysiąclecia Państwa Polskiego 7, 25-314 Kielce, Poland

tmpmj@tu.kielce.pl

received, revised, accepted

Abstract: Temporal measurements obtained during flow boiling experiments are strongly nested within individual experimental runs, which complicates validation of data-driven heat transfer models. This study examines temporal characteristics of prediction errors from Random Forest (RF) models for local Nusselt number Nu and heat transfer coefficient α . Rather than developing a new heat transfer model, the analysis investigates whether temporal descriptors can identify conditions associated with larger RF errors and whether these associations transfer to independent experiments. The analysis was restricted to the operational subcooled branch to preserve one thermal-reference formulation. The archive covered six dielectric working fluids, several asymmetrically heated minichannel configurations, smooth and enhanced surfaces, varying thermal-flow conditions and module orientations. The minichannel depth was 1 mm and the observed length 43 mm; infrared thermography provided the heated foil surface temperature. After final metadata quality control, the corrected 1 Hz cohort comprised 434 experiments, 254,152 profiles and 6,511 temporal segments, of which 5,581 formed the principal quality-controlled cohort. Residual magnitude and signed bias were related to 181 temporal descriptors using experiment-clustered, train-only inference. The corrected screening retained 55 and 58 error-magnitude associations for Nu and α , and 14 and 16 signed-bias associations, respectively. Signed-bias directions were fully preserved in validation and test, whereas error-magnitude stability was weaker, especially for α in test. For nonlinear residual models using all 181 descriptors, random segment-level cross-validation produced apparent MAE reductions of 29.6% for Nu and 31.2% for α relative to experiment-grouped validation. On the test experiments, none of the residual-prediction models outperformed the median baseline, indicating that apparent segment-level predictive skill did not transfer to unseen experiments. The proposed experiment-aware residual framework can therefore identify conditions associated with larger model errors, detect overly optimistic validation caused by within-experiment information sharing, and support model refinement and targeted data acquisition in densely sampled flow boiling experiments.

Key words: flow boiling; rectangular minichannel; Random Forest; temporal descriptors; experiment-grouped validation; data leakage; residual analysis; Nusselt number; heat transfer coefficient

1. INTRODUCTION

Flow boiling in mini- and minichannels is intrinsically dynamic. Wall temperature, pressure, mass flow, heat flux and two-phase structure may evolve jointly along an experimental trajectory, while oscillatory behaviour and regime transitions depend strongly on the hydraulic system and operating point [1-4]. In the present work, this temporal variability is treated as a property of the experimental trajectory; the available signals are not used to assign a particular hydrodynamic-instability mechanism.

A complementary line of research has addressed the reconstruction and computational verification of local heat transfer quantities in time-dependent minichannel experiments. Maciejewska and Piasecka [5] formulated an unsteady inverse thermal solution based on Trefftz functions, while subsequent studies quantified the propagation of temperature-measurement uncertainty and compared thermographic methods [6,7]. Of particular relevance, Pawińska et al. [8] combined the homotopy perturbation method with Trefftz functions and Simcenter STAR-CCM+ for the analysis of flow boiling heat transfer. The present study continues this computational and verification line from a different direction: the

reconstructed heat transfer quantities are treated as fixed experimental outputs, and the temporal behaviour of prediction errors from a data-driven model is examined.

Machine-learning studies of flow boiling have demonstrated high interpolation accuracy, but they also reveal that predictive difficulty depends on database composition and the physical domain represented during training. The recent review by Chu et al. [9] identifies uneven datasets, overfitting and weak extrapolation as recurring limitations in boiling applications. Qiu et al. [10] trained an artificial neural network on 16,953 saturated flow boiling records from 50 sources and 16 working fluids, obtaining a test MAE of 14.3%; when complete databases were withheld, prediction deteriorated markedly when the corresponding working fluid was absent from training. Bard et al. [11] reanalysed the same consolidated database with seven model families and extensive feature-selection procedures; support-vector regression gave the best overall MAPE of 11.3%, while the exploratory analysis also exposed strong imbalance and influential extreme observations. More recent flow boiling studies extended the model set to XGBoost and multi-model physics-aligned surrogates [12,13]. Mudawar et al. [14] considered 29,226 heat transfer records spanning highly subcooled to high-quality saturated boiling under microgravity and different Earth-gravity orientations; their separate neural networks predicted the

subcooled and saturated subsets with MAEs of 6.31% and 11.70%, respectively, illustrating that the two thermal regimes need not have the same predictive difficulty.

The validation strategy is therefore inseparable from the model result. Loyola-Fuentes et al. [15] proposed a six-stage framework for regression of heat transfer data using machine learning, explicitly including data pretreatment, feature selection, the splitting philosophy, model training, hyperparameter tuning and performance assessment. In their case study, machine learning improved interpolation but extrapolated less reliably than a semi-empirical correlation. A related robustness study by Mehdi et al. [16] reported distinctly larger errors on deliberately excluded fluid-surface datasets than on a conventional train/test split, while Wang and Kharangate [17] showed that database size alone does not determine information content and that prediction uncertainty is strongly linked to coverage of the input domain. These results reinforce the distinction between interpolation accuracy and transfer to an unseen experiment or physical domain.

Dense experimental archives create an additional validation problem because many local points, profiles or temporal segments may originate from a much smaller number of physical experiments. If related observations from one experiment occur on both sides of a train-validation boundary, the reported error can be artificially reduced [18]. For temporal and hierarchical data, blocking or grouping should represent the intended transfer task rather than merely randomise records [19,20]. This issue is central here because segments from one experiment share geometry, calibration, operating trajectory and data-reduction assumptions.

Residual analysis is also related to, but distinct from, change-point detection and non-stationary process monitoring [21-25]. Those approaches are designed to detect persistent changes, anomalies or evolving process states and, in some cases, to adapt a model online. Here the temporal descriptors are used retrospectively: the objective is to determine whether residual associations identified in training experiments recur in physically independent experiments. No online change-point detector, drift-adaptation algorithm or hydrodynamic-instability classifier is introduced.

The immediate methodological foundation of the present work is the source-file-level pilot benchmark based on Random Forests [26] and the integrated database documented in the preprint by Piasecka and Piasecki [27]. This article describes the rectangular-minichannel database, the harmonisation and quality-control workflow, operational regime labelling and leakage-safe source-file separation. Under that separation, the pilot RF reached R^2 values of 0.88 for subcooled Nu and 0.80 for subcooled α , compared with 0.46 and 0.41 for the corresponding saturated targets. The current residual study was defined after that benchmark and deliberately restricted to the operational subcooled branch. This restriction keeps one reference-temperature formulation throughout the temporal analysis and uses frozen RF models with sufficient predictive signal for meaningful residual diagnostics, rather than mixing errors produced under different branch-specific thermal definitions. The article [27] is particularly relevant here because it documents the integrated data harmonisation, operational regime definition and leakage-safe source-file separation on which the present temporal-residual analysis is built.

Accordingly, the present study examines temporal characteristics of residual magnitude and signed bias produced by RF models for the operational subcooled branch. Associations identified using training experiments are evaluated in independent validation and test experiments, while multivariable residual models are assessed under experiment-grouped cross-validation and compared directly

with random segment-level validation. The analysis is deliberately confined to the subcooled branch to retain a single thermal-reference definition and a physically coherent target domain. Application of the same experiment-aware protocol to the saturated branch is reserved for a subsequent regime-to-regime comparison.

2. MATERIALS AND METHODS

2.1. Experimental Archive, Measurement Chain and Thermal Reconstruction

The analysis used historical rectangular-minichannel flow boiling experiments conducted at Kielce University of Technology. The experimental programme used closed-loop facilities comprising a working-fluid circuit, electrical heating and control, and synchronised thermal and visual measurement systems. The general measurement chain included a circulation pump, pressure-stabilisation and heat-exchange components, a Coriolis mass flowmeter, inlet and outlet pressure and fluid-temperature measurements, and current-voltage measurements defining the electrical heat input. Infrared thermography observed the external surface of the thin asymmetrically heated metal plate, while the opposite side of representative modules was optically accessible through glass for high-speed flow visualisation [5-8,27]. Because the archive combines several campaigns and modular test sections, fluid, geometry, heated-surface condition and orientation were retained as experimental provenance rather than reduced to a single nominal configuration.

The experimental archive covered several modular rectangular-minichannel configurations with different numbers of parallel, asymmetrically heated channels. The minichannel depth was fixed at 1 mm and the observed channel length at 43 mm, whereas the channel arrangement and width varied between test modules. The experiments were performed under different thermal-flow conditions, with smooth and enhanced heated surfaces, different spatial orientations, and six dielectric working fluids circulated in the closed loop: FC-72, FC-770, HFE-649, HFE-7000, HFE-7100 and HFE-7200. Across these campaigns, the outer-surface temperature of the heated foil was obtained by infrared thermography.

For each experiment, the central longitudinal infrared wall-temperature profile was associated with the corresponding operating conditions and geometric metadata. Local pressure and a stream-wise fluid-temperature reference were reconstructed from inlet and outlet measurements. The present residual analysis uses only the operational subcooled branch; within this branch, the local heat transfer coefficient α and Nusselt number Nu were evaluated from Eqs. (1)-(3). External heat losses and conduction through the heated plate were treated according to the established thermal reconstruction [5-8,27].

The heat transfer quantities were evaluated from:

$$\alpha = \frac{q_{net}}{\Delta T_{eff}} \quad (1)$$

$$\Delta T_{eff} = T_w - T_{f,ref} - \Delta T_{cond} \quad (2)$$

$$Nu = \frac{\alpha D_h}{k_f} \quad (3)$$

where: α is the local heat transfer coefficient, q_{net} is the local net heat flux transferred to the fluid after heat-loss correction, ΔT_{eff} is the effective wall-to-fluid temperature difference, T_w is the local heated-wall temperature, $T_{f,ref}$ is the fluid reference temperature

used by the operational branch, and ΔT_{cond} is the temperature correction for conduction through the heated plate. D_h denotes the hydraulic diameter and k_f the thermal conductivity of the working fluid. The net heat flux q_{net} is therefore not the raw electrical heat input; it is the portion assigned to heat transfer to the fluid after the adopted heat-loss correction.

Subcooled flow boiling is understood here at the experiment level: the bulk or reference liquid remains below the local saturation temperature, while vapour generation may occur at the heated surface during the flow-boiling run. For reproducible data reduction, the integrated database follows the operational convention established in [27]. The sign of $\Delta T_{ws} = T_w - T_{sat}$ is used to select the thermal-reference branch; it is not used as an onset-of-nucleate-boiling criterion for an individual local record. Records with $\Delta T_{ws} \leq -2$ K are assigned to the operational subcooled-boiling branch and use the linearly reconstructed bulk-fluid temperature $T_{f,lin}$ as $T_{f,ref}$. Records with $\Delta T_{ws} \geq +2$ K are assigned to the saturated-reference branch and use T_{sat} . The symmetric deadband $|\Delta T_{ws}| < 2$ K is retained as an ambiguous branch and excluded from α and Nu calculation. Thus, the term “subcooled boiling region” in this article denotes the operational subcooled-reference branch of the flow-boiling archive, consistent with the preceding data reduction, and does not imply that $\Delta T_{ws} < 0$ itself proves local boiling at the wall or constitutes a complete flow-pattern classification.

The present article is intentionally restricted to the operational subcooled boiling region. The restriction avoids combining residuals based on different reference-temperature definitions and follows the preceding source-file-level benchmark, in which the frozen RF models were substantially more informative for the subcooled targets than for the saturated targets [27]. The saturated boiling region will be analysed separately using the same experiment-aware residual framework in a future work.

High-speed visualisation provided physical context in the original campaigns but was not available as a harmonised temporal predictor for the present analysis. Consequently, the statistical descriptors below cannot by themselves assign segments to a named flow pattern, density-wave oscillation, pressure-drop oscillation, Ledinegg instability (static flow excursion) or dryout occurrence.

2.2. Temporal Cohort and Quality Control

The corrected 1 Hz reconstruction used in this study covered 434 experiments and yielded 254,152 complete profiles grouped into 6,511 temporal segments. During final provenance quality control, one training experiment was excluded because its working-fluid identifier was confirmed to be inconsistent with the source-file identity; no validation or test experiment was affected. Sequence, experimental provenance and the predefined processing boundaries were otherwise preserved. Primary RF prediction residuals were available for 5,947 temporal segments, and 5,581 of these also met the predefined quality-control acceptance criteria and formed the principal analysis cohort. These populations are summarised in Tab. 1.

Within each source file, complete profiles were ordered by the preserved acquisition sequence. A new temporal segment was started at a source-file boundary, a break in the reconstructed one-sample-per-second sequence, a change in the operational branch, or a predefined processing boundary inherited from the upstream reconstruction. The latter boundaries were fixed before RF residuals were generated and were not selected from prediction-error behaviour. Each segment therefore represented one contiguous,

provenance-preserving interval and retained its source-file and experiment identifiers. Segment sample count and duration were retained explicitly in the frozen temporal metadata; no additional segment splitting or merging was introduced during the residual analysis. The principal cohort retained only segments accepted by the frozen multichannel quality-control protocol.

Tab. 1. Temporal cohort used in the residual analysis.

Population	Experiments	Profiles	Segments
1 Hz temporal cohort	434	254,152	6,511
RF residuals available	434	217,916	5,947
Principal analysis	434	-	5,581
Train / validation / test	304 / 65 / 65	-	3,872 / 859 / 850

When analysing the data listed in Tab. 1, it can be seen that the principal analysis comprised 5,581 segments nested within 434 experiments. The experiment-grouped split contained 304 training, 65 validation and 65 test experiments, contributing 3,872, 859 and 850 principal-cohort segments, respectively. Segment observations from the same experiment share geometry, calibration, operating trajectory and data-reduction assumptions; they therefore cannot be treated as independent experimental replicates. The experiment-aware analysis sequence used here is shown in Fig. 1.

Fig. 1 illustrates how the successive processing stages reduced the 6,511 reconstructed temporal segments to 5,581 QC-eligible segments while preserving the experiment-level structure. Consequently, the experiment, rather than an individual temporal segment, was retained as the final validation unit.

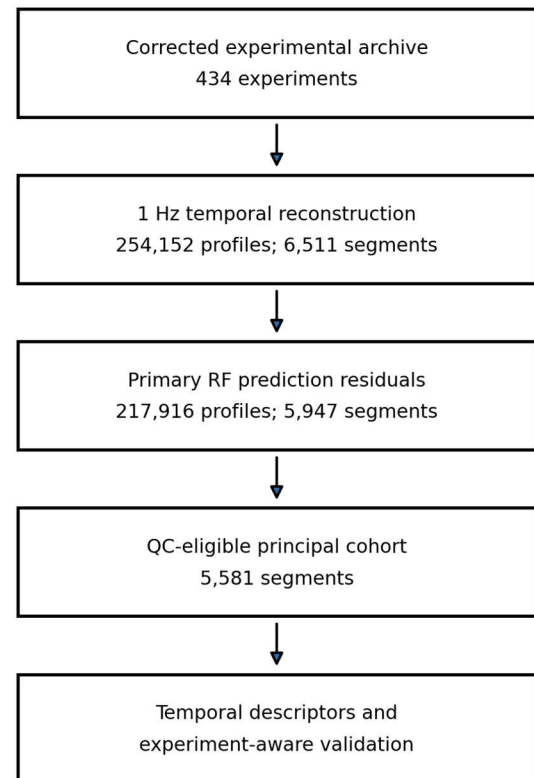


Fig. 1. Analysis workflow for the corrected cohort, from the experimental archive to segment-level residual association and experiment-aware validation.

2.3. Random Forest Residuals and Segment-Level Outcomes

Predictions were generated by saved-model inference using the RF regressors defined in the preceding database benchmark [27]. The RF architecture, feature formulation and hyperparameters were retained unchanged, and no new RF training or hyperparameter retuning was performed in the present temporal-residual analysis. After the final provenance correction described in Section 2.2, the saved models were applied to the corrected cohort to regenerate predictions and residuals. The reference formulation predicts local Nu and α without the wall heat-flux predictor q_w , which is closely linked to the thermal reconstruction. Residuals were defined as prediction minus observation. Local residuals were first aggregated within each complete thermographic profile and then within temporal segments, preventing long spatial profiles from receiving disproportionate statistical weight.

The preceding benchmark [27] selected the no_q_w formulation for subcooled Nu and the full formulation for subcooled α as its preferred pilot configurations. For the present residual study, no_q_w was adopted as the common primary formulation for both targets so that the two residual families used one controlled upstream input definition. The common input set comprised Re , Pr , D_h , q_{net} , T_{sat} , $T_{f,lin}$, $T_{f,used}$, and θ . Thus, only the applied wall heat flux q_w was removed; q_{net} remained available and, for α , is directly related to the target definition. The no_q_w choice should therefore be interpreted as a controlled common input configuration, not as removal of all target-proximal information. The archived full-input formulation differs by including q_w and did not define the primary residuals reported here.

For the frozen primary no_q_w RF models, the performance metrics archived in the original baseline-run *metrics.csv* files were calculated on the source-file-grouped point-level test sets (300,000 local records per target): $R^2 = 0.794$, $RMSE = 3.083$, $MAE = 1.966$ and $bias = 0.345$ for Nu , and $R^2 = 0.663$, $RMSE = 139.437 \text{ W m}^{-2} \text{ K}^{-1}$, $MAE = 90.372 \text{ W m}^{-2} \text{ K}^{-1}$ and $bias = 23.453 \text{ W m}^{-2} \text{ K}^{-1}$ for α . These values characterize the original RF test partitions and are not recomputed performance metrics for the provenance-corrected AMA cohort; the residuals analysed below were generated by applying the saved models to the corrected profile-point data and were then aggregated to profile and segment level.

Residuals were calculated at each local profile point. Within each complete thermographic profile, the mean residual defined profile bias and the mean absolute residual defined profile MAE. Segment-level bias and MAE were then obtained by averaging the corresponding profile-level quantities, so each complete profile contributed equally regardless of the number of retained spatial points.

Residual magnitude was represented by a monotonic logarithmic transformation of a non-negative error measure E , consistent with standard transformation practice for positively skewed variables [28]: $M_E = \ln(1+E/E_0)$. The reference quantity E_0 renders the logarithm argument dimensionless and was set to 1 for Nu and $1 \text{ W m}^{-2} \text{ K}^{-1}$ for α . This transformation compresses the upper range while preserving the ordering of observations. The principal magnitude outcome was based on mean profile MAE and is denoted M_{MAE} ; signed mean profile bias was analysed without transformation.

2.4. Temporal Descriptors

The temporal descriptors were derived solely from measured or reconstructed experimental signals and did not include RF residuals or residual-derived quantities. The principal set comprised 181 descriptors summarising signal level, dispersion, trend and short-term variability within each segment. These included minimum, maximum and quantile values; measures of dispersion and detrended dispersion; initial and final values and net change; linear slope; step-to-step variability and the fraction of sign changes between successive steps; and autocorrelation between values one sample apart, corresponding to a 1-s shift at the 1 Hz reconstruction frequency. The descriptors were calculated for fluid temperature, wall temperature, mass flow, applied and net heat fluxes, the wall-to-fluid temperature difference, and the difference between applied and net heat flux. The four principal residual families are summarised in Tab. 2.

Tab. 2. Residual outcomes and temporal predictor sets.

Target / residual outcome	Interpretation	Predictors
Nu : M_{MAE}	error magnitude	181 temporal descriptors
Nu : mean profile bias	signed residual	
α : M_{MAE}	error magnitude	
α : mean profile bias	signed residual	

Tab. 2 distinguishes two complementary residual outcomes. The transformed error magnitude describes conditions associated with larger or smaller RF errors, whereas signed bias indicates systematic over- or under-prediction. Bias in α is dimensional and bias in Nu is not; the two targets are therefore interpreted separately. The logarithmic transformation is monotonic and consequently does not alter the rank ordering used in the Spearman analysis.

2.5. Association Analysis and Cross-Experiment Stability

Spearman rank correlation was used to quantify monotonic associations between individual temporal descriptors and each residual outcome [29]. The coefficient ρ ranges from -1 to +1; its sign gives the direction of the association and its absolute value gives the strength of the rank relation, without requiring linearity. All inferential screening was performed using training experiments only. Uncertainty was evaluated with 1,000 bootstrap resamples clustered by experiment, and rank-regression p-values were calculated with experiment-cluster-robust standard errors. Benjamini-Hochberg control was applied at a false-discovery rate of 0.05. An association was retained when the adjusted significance criterion was met, the 95% cluster-bootstrap interval excluded zero and $|\rho|$ was at least 0.10. The Benjamini-Hochberg adjustment was applied separately within each of the four residual families across its 181 train-only descriptor tests.

All 181 descriptors were screened within each residual family using training experiments only. The corrected analysis retained 143 train discoveries in total. All 143 were carried forward unchanged for validation- and test-set stability assessment. Separately, to reduce redundancy for the compact multivariable models, train discoveries were ordered by decreasing absolute Spearman correlation and greedily pruned when their pairwise absolute rank correlation with an already retained descriptor was 0.90 or greater,

with at most 12 descriptors per family. This produced 34 non-redundant descriptors across the four family-specific compact panels (12, 5, 12 and 5). Associations in the held-out validation and test experiments were classified as same direction (S), effectively null when $|p| < 0.05$ (N), or sign reversal (R).

2.6. Multivariable Residual Models and Validation Design

To determine whether temporal descriptors jointly explain residual variation, three segment-level models were considered: a median baseline, elastic-net regression with fold-specific standardisation, and histogram-based gradient boosting (HGB). These models describe residuals of the RF heat transfer model; they are not competing heat transfer models. Both the complete 181-descriptor set and the corrected family-specific compact panels were examined. Residual-model hyperparameters had been selected from the training partition by five-fold cross-validation grouped by experiment, with MAE as the principal criterion. After the final provenance correction, those hyperparameters were frozen and reevaluated without retuning; the compact predictor panels alone were rebuilt from the corrected train-only association screen using the unchanged redundancy rule.

The effect of the validation unit was evaluated using a second five-fold scheme in which temporal segments, rather than experiments, were randomly assigned to folds. Identical frozen model configurations were used in the grouped and random-segment schemes, so the comparison isolates the apparent performance gain caused by allowing segments from the same experiment to occur in both fitting and validation folds. Validation and test experiments were not used for predictor selection, hyperparameter tuning or model choice. After the grouped and random-segment comparison, the frozen configurations were fitted to the complete corrected training subset and evaluated on the locked validation and test subsets. Apparent MAE reduction was defined as $100 \times [\text{MAE}(\text{grouped}) - \text{MAE}(\text{random})] / \text{MAE}(\text{grouped})$; positive values therefore quantify the optimism introduced by random segment-level validation.

3. RESULTS AND DISCUSSION

3.1. Temporal Associations and Cross-Experiment Stability

The corrected training analysis identified 55 significant temporal associations with Nu error magnitude and 58 with α error magnitude. Signed bias yielded fewer associations, 14 for Nu and 16 for α . The maximum absolute training-set Spearman coefficient was 0.219 for Nu error magnitude, 0.146 for Nu bias, 0.236 for α error magnitude and 0.143 for α bias. The two highest-ranked train-selected descriptors from each residual family are reported in Tab. 3; aggregate stability counts are summarised in Tab. 4, while Fig. 2 presents the same stability counts graphically.

Tab. 3. Representative train-selected temporal descriptors and their Spearman coefficients in held-out validation and test experiments.

Family	Representative descriptor	Val. ρ	Test ρ
Nu error magnitude	ΔT_{wf} : minimum	-0.112	-0.031
Nu error magnitude	ΔT_{wf} : 5th percentile	-0.115	-0.020

Nu bias	T_f : detrended standard deviation	0.143	0.181
Nu bias	T_f : range	0.148	0.181
α error magnitude	ΔT_{wf} : minimum	-0.150	-0.065
α error magnitude	ΔT_{wf} : 5th percentile	-0.151	-0.055
α bias	T_f : step standard deviation	0.130	0.198
α bias	T_f : RMS step	0.130	0.199

When analysing the representative descriptors listed in Tab. 3, the held-out coefficients showed a clear difference between error magnitude and signed bias. For Nu error magnitude, the minimum and 5th percentile of ΔT_{wf} were still negative in validation ($\rho = -0.112$ and -0.115) but became effectively null in test (-0.031 and -0.020). For α error magnitude, the same descriptors retained negative associations in both held-out subsets ($-0.150/-0.065$ and $-0.151/-0.055$ for validation/test, respectively). In contrast, Nu -bias descriptors based on T_f detrended standard deviation and range remained positive (0.143/0.181 and 0.148/0.181), as did α -bias descriptors based on T_f step standard deviation and RMS step (0.130/0.198 and 0.130/0.199). The broader retained sets additionally included wall-temperature variability and heat-flux trajectory measures such as net change in q_{net} ; these associations identify signal families associated with model vulnerability and are not interpreted as a causal ranking of boiling mechanisms.

Signed-bias associations were more directionally stable than error-magnitude associations (Tab. 4; Fig. 2). For Nu error magnitude, 42 of 55 associations retained their direction in validation and 41 in test; 13 were effectively null and one reversed sign in test. For α error magnitude, 47 of 58 associations retained their direction in validation, but only 28 did so in test, where 29 became effectively null and one reversed sign. In contrast, every signed-bias discovery retained its direction in both held-out subsets (14/14 for Nu and 16/16 for α). The associations therefore remained directionally stable for bias, while error-magnitude structure transferred less consistently across independent experiments.

Tab. 4. Cross-experiment stability of the corrected train-selected temporal associations. S - same direction; N - effectively null; R - sign reversal.

Family	Train associations	Validation S/N/R	Test S/N/R
Nu error magnitude	55	42 / 13 / 0	41 / 13 / 1
Nu bias	14	14 / 0 / 0	14 / 0 / 0
α error magnitude	58	47 / 11 / 0	28 / 29 / 1
α bias	16	16 / 0 / 0	16 / 0 / 0

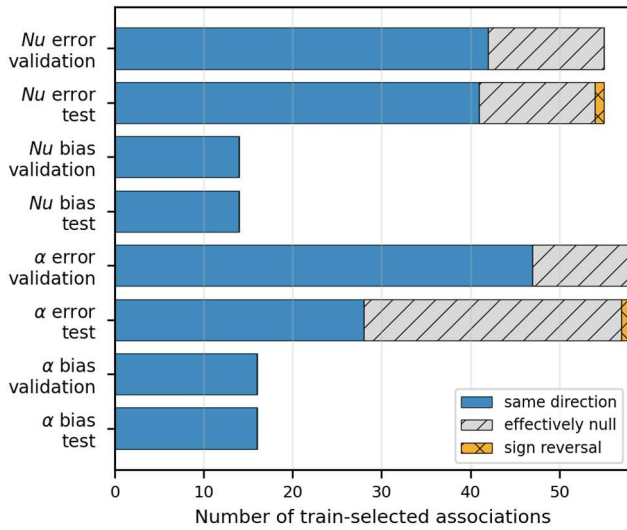


Fig. 2. Cross-experiment stability of corrected train-selected temporal associations. Blue - same direction; grey - effectively null; yellow - sign reversal.

Taken together, Tab. 4 and Fig. 2 show complete directional stability of signed bias in both held-out subsets, whereas the error-magnitude associations transfer less consistently, particularly for α in the test experiments. The temporal structure of residual sign is therefore more stable across independent experiments than the temporal structure of residual magnitude.

3.2. Effect of the Validation Scheme

The validation scheme had a major influence on apparent model performance. When segments from one experiment were allowed to occur in both fitting and validation folds, the residual models benefited from shared experimental context. Tab. 5 compares experiment-grouped and random-segment results for identical frozen HGB configurations, and Fig. 3 shows the corresponding point estimates of apparent MAE reduction.

Tab. 5. Effect of random-segment validation on corrected HGB performance.

Family / predictor set	Grouped / random R^2	Apparent MAE reduction, %
Nu error / reduced panel	0.040 / 0.270	12.1
Nu error / 181 descriptors	0.013 / 0.479	29.6
Nu bias / reduced panel	-0.091 / -0.091	0.5
Nu bias / 181 descriptors	0.059 / 0.222	12.7
α error / reduced panel	0.069 / 0.251	10.2
α error / 181 descriptors	0.024 / 0.519	31.2
α bias / reduced panel	-0.078 / -0.057	1.3
α bias / 181 descriptors	0.085 / 0.254	16.9

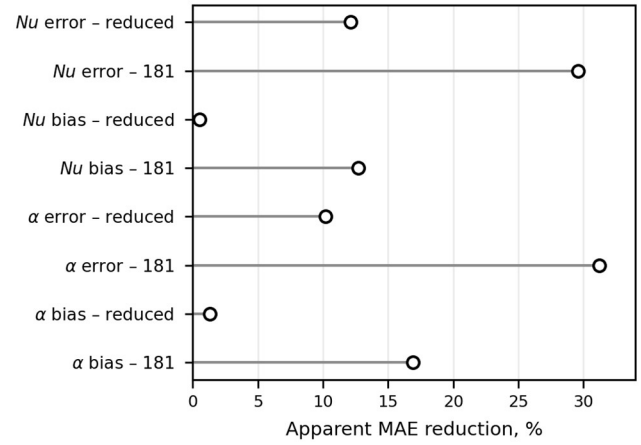


Fig. 3. Apparent reduction in MAE obtained with random-segment validation relative to experiment-grouped validation for the corrected HGB configurations. Points denote corrected estimates; the same frozen model specification was used in both validation schemes.

When analysing the results listed in Tab. 5 and shown in Fig. 3, it was observed that the largest differences occurred for error-magnitude models using all 181 descriptors. For Nu , random-segment R^2 was 0.479 compared with 0.013 under experiment-grouped validation; for α , the corresponding values were 0.519 and 0.024. Random-segment validation reduced the apparent MAE by 29.6% for Nu and 31.2% for α . The reduced error-magnitude panels still showed apparent MAE reductions of 12.1% and 10.2%, respectively. Bias families showed smaller but still notable optimism for the full 181-descriptor models (12.7% for Nu and 16.9% for α). Thus, segment-level validation can imply predictive skill that is not reproduced when entire experiments are separated.

3.3. Locked Test Performance

The frozen residual-model configurations were fitted to the complete corrected training subset and evaluated on the locked validation and test data without further tuning. The decisive result was obtained on the test experiments: the median baseline had the lowest MAE in every residual family, and all predictive configurations had negative R^2 . Tab. 6 summarises the baseline MAE together with the lowest predictive-model MAE and the best predictive R^2 observed across the frozen elastic-net and HGB configurations. These minima are descriptive summaries only and were not used for model selection.

Tab. 6. Locked-test summary across corrected frozen residual-model configurations. Predictive minima are descriptive only; no test-based model selection was performed. Error-magnitude MAE is reported on the logarithmically transformed outcome scale; Nu bias is dimensionless, whereas α bias is reported in $W m^{-2} K^{-1}$.

Family	Median baseline MAE	Lowest predictive MAE	Best predictive R^2
Nu error magnitude	0.258	0.306	-0.016
Nu bias	0.528	0.536	-0.062
α error magnitude	0.733	0.776	-0.017
α bias	24.21	24.33	-0.023

For Nu error magnitude, the baseline MAE was 0.258, whereas the lowest predictive MAE was 0.306. The corresponding values were 0.528 versus 0.536 for Nu bias, 0.733 versus 0.776 for α error magnitude, and 24.21 versus 24.33 for α bias (Tab. 6). Even the best test R^2 among the predictive configurations remained below zero in every family. The corrected analysis therefore confirms that statistically detectable temporal associations do not provide a transferable residual-correction model for unseen experiments.

3.4. Physical Interpretation and Limitations

The corrected train-only screening retained descriptors related to wall-to-fluid temperature difference, wall-temperature variability, heat-flux evolution and fluid-temperature dynamics. These associations are physically plausible because the thermal reconstruction becomes more sensitive when the effective wall-to-fluid temperature difference is small. Since α is proportional to q_{net} and inversely proportional to ΔT_{eff} , a comparable perturbation in temperature or heat flux has a larger relative effect when ΔT_{eff} is small; Nu obtained this sensitivity through its dependence on α [5-8]. The observed associations are therefore consistent with reconstruction sensitivity and non-stationary thermal evolution, but they are not evidence of a specific boiling-instability mode.

Bias-related descriptors require similar caution. Fluid-temperature variability, autocorrelation and net changes in thermal differences can accompany slow evolution of the reference state and hence systematic residual direction. However, the historical archive does not provide a harmonised per-segment uncertainty budget or a uniform segment-level flow-pattern classification within the operational subcooled branch. The descriptors therefore cannot distinguish measurement drift, reconstruction sensitivity, hydrodynamic instability or other mechanisms with confidence. Residual magnitude is treated as an empirical model-reliability outcome, not as a calibrated measurement-uncertainty estimate.

Experiment grouping prevents direct within-experiment leakage but does not factorially balance fluid, geometry, surface and orientation. Weak cross-experiment transfer may therefore combine temporal non-stationarity, domain shift, measurement variability and reconstruction sensitivity. The 434 experiments provide the appropriate validation unit for the 5,581 correlated principal-cohort segments. A natural continuation is to apply the same frozen, experiment-aware protocol to the saturated branch and then compare the two operational branches directly while retaining their branch-specific thermal-reference definitions.

4. CONCLUSIONS

This study analysed temporal characteristics of Random Forest prediction errors for local Nusselt numbers and heat transfer coefficients in a heterogeneous archive of rectangular-minichannel flow-boiling experiments spanning six dielectric working fluids, multiple channel configurations, surface states, thermal-flow conditions and module orientations. The corrected 1 Hz cohort comprised 434 experiments and 6,511 temporal segments, of which 5,581 residual-covered segments met the principal quality-control criteria. Experiment identity, rather than an individual temporal segment, was retained as the validation unit.

The principal cohort contained 304 training, 65 validation and 65 test experiments, contributing 3,872, 859 and 850 segments, respectively. This separation preserved complete experiments across

all validation stages and prevented direct within-experiment leakage.

Significant but modest temporal associations were found for both Nu and α , with maximum $|\rho|$ not exceeding 0.236. The corrected train-only screen retained 55 Nu error-magnitude, 14 Nu bias, 58 α error-magnitude and 16 α bias associations. Signed-bias directions were preserved in both held-out subsets, whereas error-magnitude associations transferred less consistently, particularly for α in test.

The temporal descriptors did not support reliable residual correction in unseen experiments. Random segment-level validation produced apparent MAE reductions of 29.6% for Nu and 31.2% for α error magnitude when all 181 descriptors were used, whereas experiment-grouped validation gave much lower R^2 values. On the locked test experiments, the median baseline had the lowest MAE in all four residual families and every predictive configuration had negative R^2 . This demonstrates the magnitude of within-experiment information sharing when densely sampled segments are treated as if they were independent.

Temporal error characteristics are therefore useful for identifying operating conditions associated with larger model errors, but they do not by themselves constitute a transferable correction model or provide a diagnosis of a specific boiling instability. The experiment-aware residual framework can instead support the assessment of model reliability, reveal overly optimistic validation caused by within-experiment information sharing, and indicate where additional data or model refinement may be needed. The next stage should apply the same protocol to the saturated branch and compare branch-specific residual behaviour without mixing the underlying thermal-reference definitions.

NOMENCLATURE

α	– local heat transfer coefficient, $W\ m^{-2}\ K^{-1}$
Nu	– Nusselt number, –
q_{net}	– local net heat flux transferred to the fluid after heat-loss correction, $W\ m^{-2}$
T_w	– wall temperature, $^{\circ}C$
$T_{f,ref}$	– fluid reference temperature, $^{\circ}C$
ΔT_{eff}	– effective wall-to-fluid temperature difference, K
ΔT_{cond}	– heated-plate conduction correction, K
D_h	– hydraulic diameter, m
k_f	– fluid thermal conductivity, $W\ m^{-1}\ K^{-1}$
ρ	– Spearman rank correlation coefficient, –
R^2	– coefficient of determination, –
q_w	– applied wall heat flux, $W\ m^{-2}$
$T_{f,lin}$	– local bulk-fluid temperature, $^{\circ}C$
T_{sat}	– saturation temperature, $^{\circ}C$
ΔT_{ws}	– wall-to-saturation temperature difference, K
T_f	– fluid temperature used in temporal descriptors, $^{\circ}C$
ΔT_{wf}	– wall-to-fluid temperature difference, $T_w - T_f$, K
E	– non-negative residual error measure, – or $W\ m^{-2}\ K^{-1}$
E_0	– reference quantity used to nondimensionalise E , – or $W\ m^{-2}\ K^{-1}$
M_E	– logarithmically transformed error magnitude, –
M_{MAE}	– transformed mean-profile MAE, –

Abbreviations

RF	– Random Forest
HGB	– histogram-based gradient boosting
MAE	– mean absolute error
RMSE	– root-mean-square error
IQR	– interquartile range

CV	– cross-validation
FDR	– false discovery rate
QC	– quality control
CI	– confidence interval
S/N/R	– same direction / null / sign reversal

REFERENCES

- Wang B, Hu Y, He Y, Rodionov N, Zhu J. Dynamic instabilities of flow boiling in micro-channels: A review. *Applied Thermal Engineering*. 2022;214:118773. <https://doi.org/10.1016/j.applthermaleng.2022.118773>.
- Prajapati YK, Bhandari P. Flow boiling instabilities in microchannels and their promising solutions - A review. *Experimental Thermal and Fluid Science*. 2017;88:576-593. <https://doi.org/10.1016/j.expthermflusci.2017.07.014>.
- Grzybowski H, Mosdorf R. Dynamics of pressure drop oscillations during flow boiling inside minichannel. *International Communications in Heat and Mass Transfer*. 2018;95:25-32. <https://doi.org/10.1016/j.icheatmasstransfer.2018.03.025>.
- Zaborowska I, Grzybowski H, Rafalko G, Mosdorf R. Boiling dynamics in parallel minichannel system with different inlet solutions. *International Journal of Heat and Mass Transfer*. 2021;165:120655. <https://doi.org/10.1016/j.ijheatmasstransfer.2020.120655>.
- Maciejewska B, Piasecka M. Trefftz function-based thermal solution of inverse problem in unsteady-state flow boiling heat transfer in a minichannel. *International Journal of Heat and Mass Transfer*. 2017;107:925-933. <https://doi.org/10.1016/j.ijheatmasstransfer.2016.11.003>.
- Piasecka M, Maciejewska B, Piasecki A. Heat Transfer Calculations during Flow in Mini-Channels with Estimation of Temperature Uncertainty Measurements. *Energies*. 2023;16:1222. <https://doi.org/10.3390/en16031222>.
- Piasecka M, Piasecki A, Maciejewska B. Liquid Crystal Thermography and Infrared Thermography Application in Heat Transfer Research on Flow Boiling in Minichannels. *Energies*. 2025;18:940. <https://doi.org/10.3390/en18040940>.
- Pawińska A, Piasecki A, Dadas N, Hożejowska S, Piasecka M. Homotopy Perturbation Method with Trefftz Functions and Simcenter STAR-CCM+ Used for the Analysis of Flow Boiling Heat Transfer. *Acta Mechanica et Automatica*. 2024;18:233-243. <https://doi.org/10.2478/ama-2024-0027>.
- Chu H, Ji T, Yu X, Liu Z, Rui Z, Xu N. Advances in the application of machine learning to boiling heat transfer: A review. *International Journal of Heat and Fluid Flow*. 2024;108:109477. <https://doi.org/10.1016/j.ijheatfluidflow.2024.109477>.
- Qiu Y, Garg D, Zhou L, Kharangate CR, Kim S-M, Mudawar I. An artificial neural network model to predict mini/micro-channels saturated flow boiling heat transfer coefficient based on universal consolidated data. *International Journal of Heat and Mass Transfer*. 2020;149:119211. <https://doi.org/10.1016/j.ijheatmasstransfer.2019.119211>.
- Bard A, Qiu Y, Kharangate CR, French R. Consolidated modeling and prediction of heat transfer coefficients for saturated flow boiling in mini/micro-channels using machine learning methods. *Applied Thermal Engineering*. 2022;210:118305. <https://doi.org/10.1016/j.applthermaleng.2022.118305>.
- Noh H, Kim J, Lee S, Kim S-M, Mudawar I. XGBoost algorithm for predicting heat transfer coefficient of saturated flow boiling in mini/micro-channels. *International Journal of Heat and Mass Transfer*. 2026;256:128095. <https://doi.org/10.1016/j.ijheatmasstransfer.2025.128095>.
- Ayegba PO, Basarkar SS, Abdul Quddus RM, Patil VN. Predicting heat transfer coefficient for saturated flow-boiling in micro/mini tubes using machine learning and annular-flow surrogates. *International Journal of Thermofluids*. 2026;34:101630. <https://doi.org/10.1016/j.ijft.2026.101630>.
- Mudawar I, Darges SJ, Devahdhanush VS. Prediction technique for flow boiling heat transfer and critical heat flux in both microgravity and Earth gravity via artificial neural networks (ANNs). *International Journal of Heat and Mass Transfer*. 2024;220:124998. <https://doi.org/10.1016/j.ijheatmasstransfer.2023.124998>.
- Loyola-Fuentes J, Nazemzadeh N, Diaz-Bejarano E, Mancin S, Colletti F. A framework for data regression of heat transfer data using machine learning. *Applied Thermal Engineering*. 2024;248:123043. <https://doi.org/10.1016/j.applthermaleng.2024.123043>.
- Mehdi S, Borumand M, Hwang G. Accurate and robust predictions of pool boiling heat transfer with micro-structured surfaces using probabilistic machine learning models. *International Journal of Heat and Mass Transfer*. 2024;226:125487. <https://doi.org/10.1016/j.ijheatmasstransfer.2024.125487>.
- Wang H, Kharangate CR. Uncertainty quantification of a flow boiling dataset for mini/micro-channels. *Applied Thermal Engineering*. 2026;287:129498. <https://doi.org/10.1016/j.applthermaleng.2025.129498>.
- Kapoor S, Narayanan A. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*. 2023;4:100804. <https://doi.org/10.1016/j.patter.2023.100804>.
- Roberts DR, Bahn V, Ciuti S, Boyce MS, Elith J, Guillerá-Arroita G, et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*. 2017;40:913-929. <https://doi.org/10.1111/ecog.02881>.
- Cerqueira V, Torgo L, Mozetič I. Evaluating time series forecasting models: an empirical study on performance estimation methods. *Machine Learning*. 2020;109:1997-2028. <https://doi.org/10.1007/s10994-020-05910-7>.
- Truong C, Oudre L, Vayatis N. Selective review of offline change point detection methods. *Signal Processing*. 2020;167:107299. <https://doi.org/10.1016/j.sigpro.2019.107299>.
- Blázquez-García A, Conde A, Mori U, Lozano JA. A Review on Outlier/Anomaly Detection in Time Series Data. *ACM Computing Surveys*. 2021;54(3):Article 56, 1-33. <https://doi.org/10.1145/3444690>.
- Qin SJ. Survey on data-driven industrial process monitoring and diagnosis. *Annual Reviews in Control*. 2012;36:220-234. <https://doi.org/10.1016/j.arcontrol.2012.09.004>.
- Zhao C. Perspectives on nonstationary process monitoring in the era of industrial artificial intelligence. *Journal of Process Control*. 2022;116:255-272. <https://doi.org/10.1016/j.jprocont.2022.06.011>.
- Gama J, Žilobaitė I, Bifet A, Pechenizkiy M, Bouchachia A. A survey on concept drift adaptation. *ACM Computing Surveys*. 2014;46(4):Article 44. <https://doi.org/10.1145/2523813>.
- Breiman L. Random Forests. *Machine Learning*. 2001;45:5-32. <https://doi.org/10.1023/A:1010933404324>.
- Piasecka M, Piasecki A. A Unified Experimental Database, Data Harmonisation Workflow and Leakage-Safe Pilot Machine Learning for Flow Boiling in Rectangular Minichannels. *Preprints*. 2026;202608.0221.v1. <https://doi.org/10.20944/preprints202608.0221.v1>.
- Box GEP, Cox DR. An analysis of transformations. *Journal of the Royal Statistical Society: Series B (Methodological)*. 1964;26(2):211-252. <https://doi.org/10.1111/j.2517-6161.1964.tb00553.x>.
- Spearman C. The Proof and Measurement of Association between Two Things. *The American Journal of Psychology*. 1904;15(1):72-101. <https://doi.org/10.2307/1412159>.

The research reported herein was funded by the National Science Centre, Poland, grant number UMO-2025/57/B/ST8/00907.

Data and code availability: The complete article-specific research package, including temporal-analysis datasets, residual summaries, validation outputs, and analysis code, will be archived in the PRACE-LAB repository of Kielce University of Technology. Access may be provided upon reasonable request to the corresponding author.

Magdalena Piasecka:  <https://orcid.org/0000-0003-3696-6213>